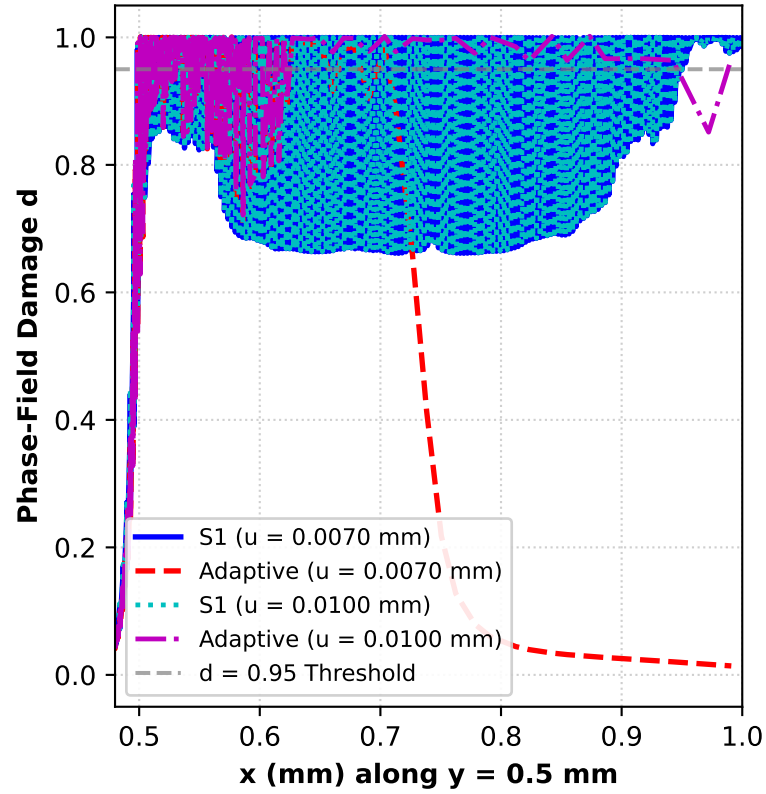
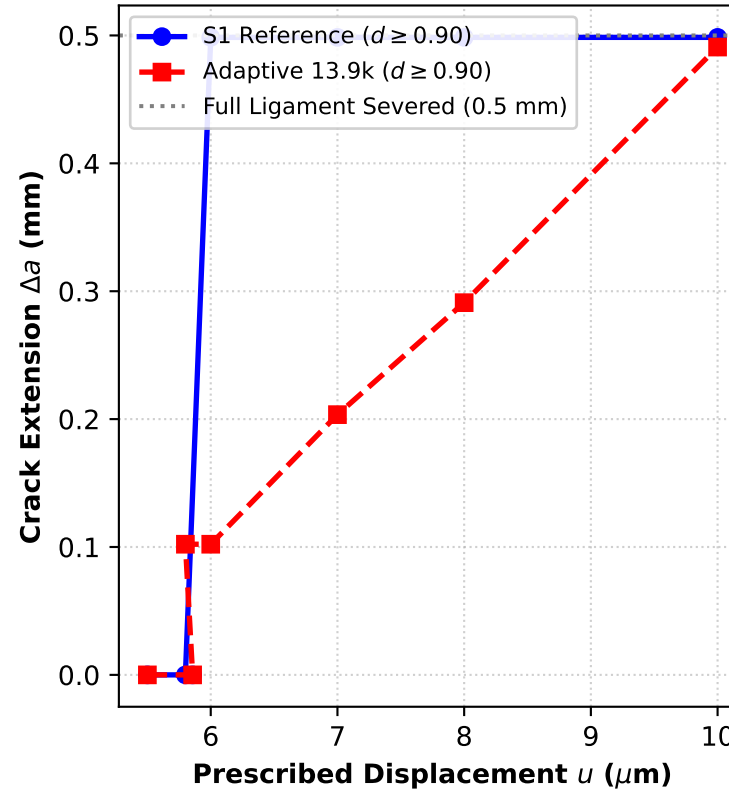


Quantitative Spatial Causality Metrics: S1 Reference vs Adaptive 13.9k Specimen

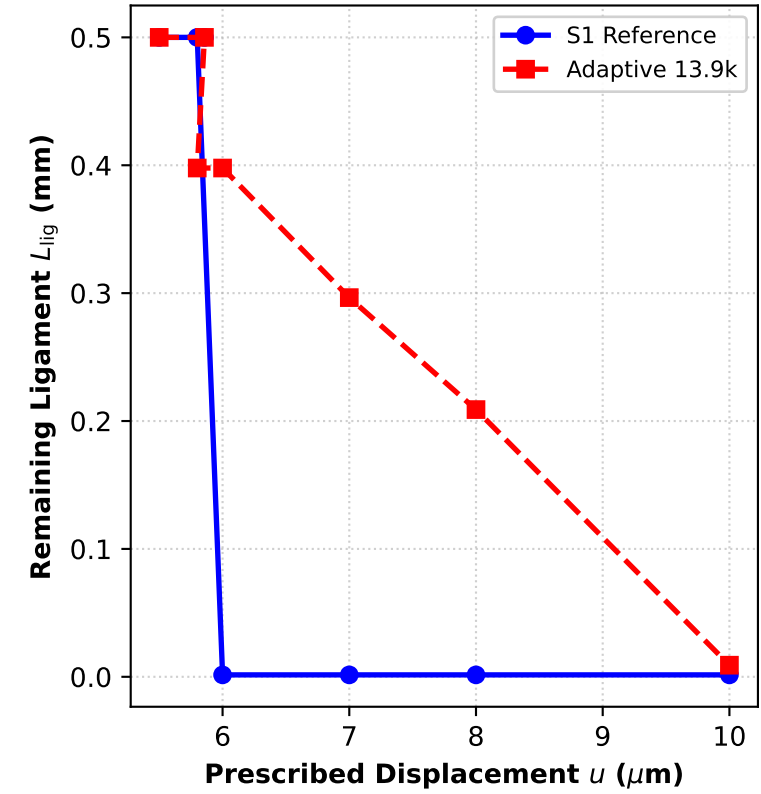
(a) Midplane Damage Profiles $d(x, y = 0.5)$



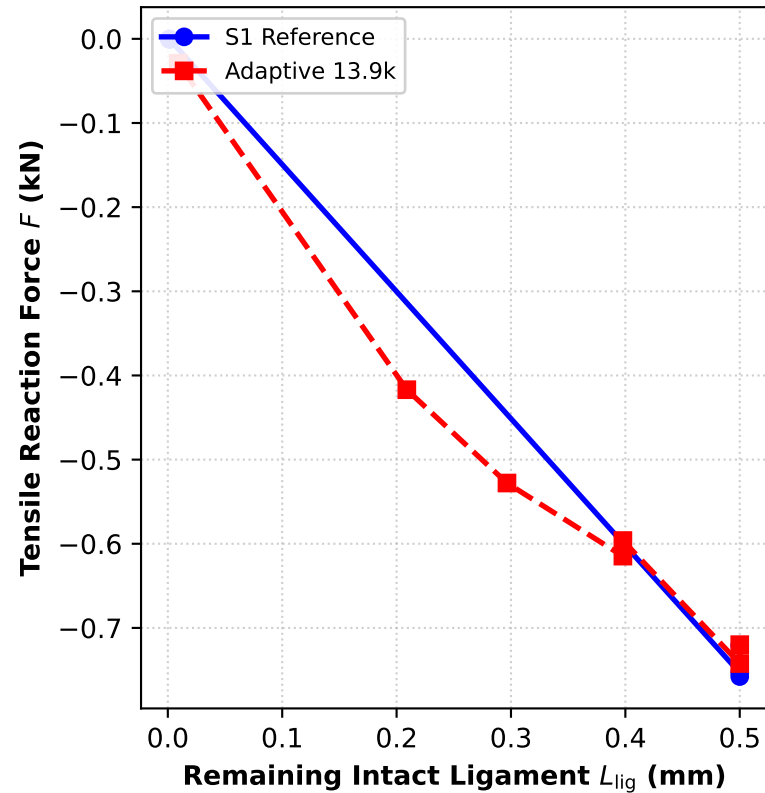
(b) Horizontal Crack Extension $\Delta a(u)$



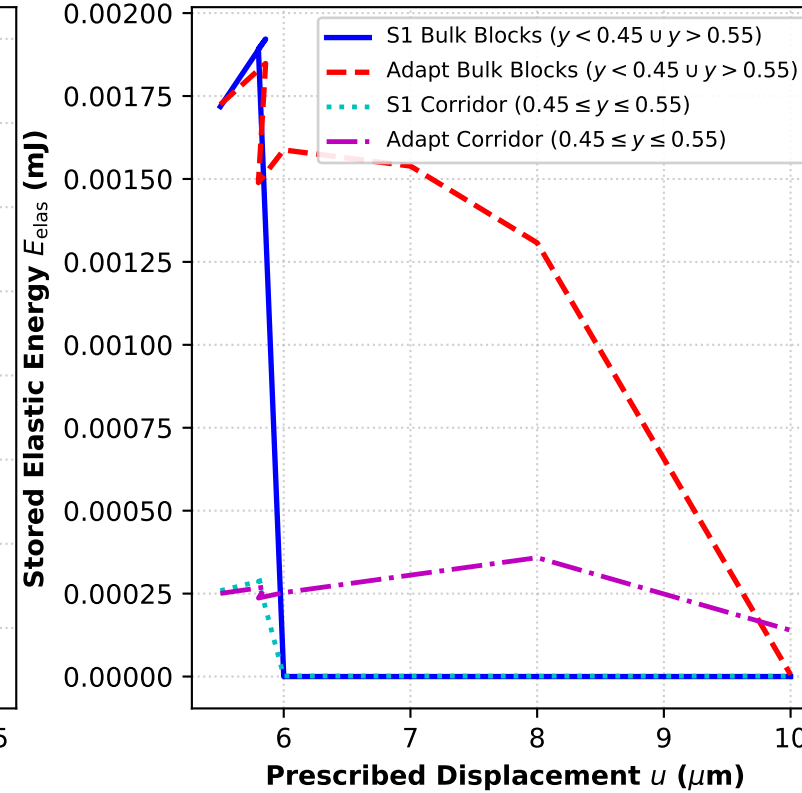
(c) Intact Ligament Length $L_{\text{lig}}(u)$



(d) Force vs Remaining Ligament $F(L_{\text{lig}})$



(e) Regional Elastic Strain Energy Partition



(f) Localization Profile $d(y)$ at $x = 0.70$ mm

